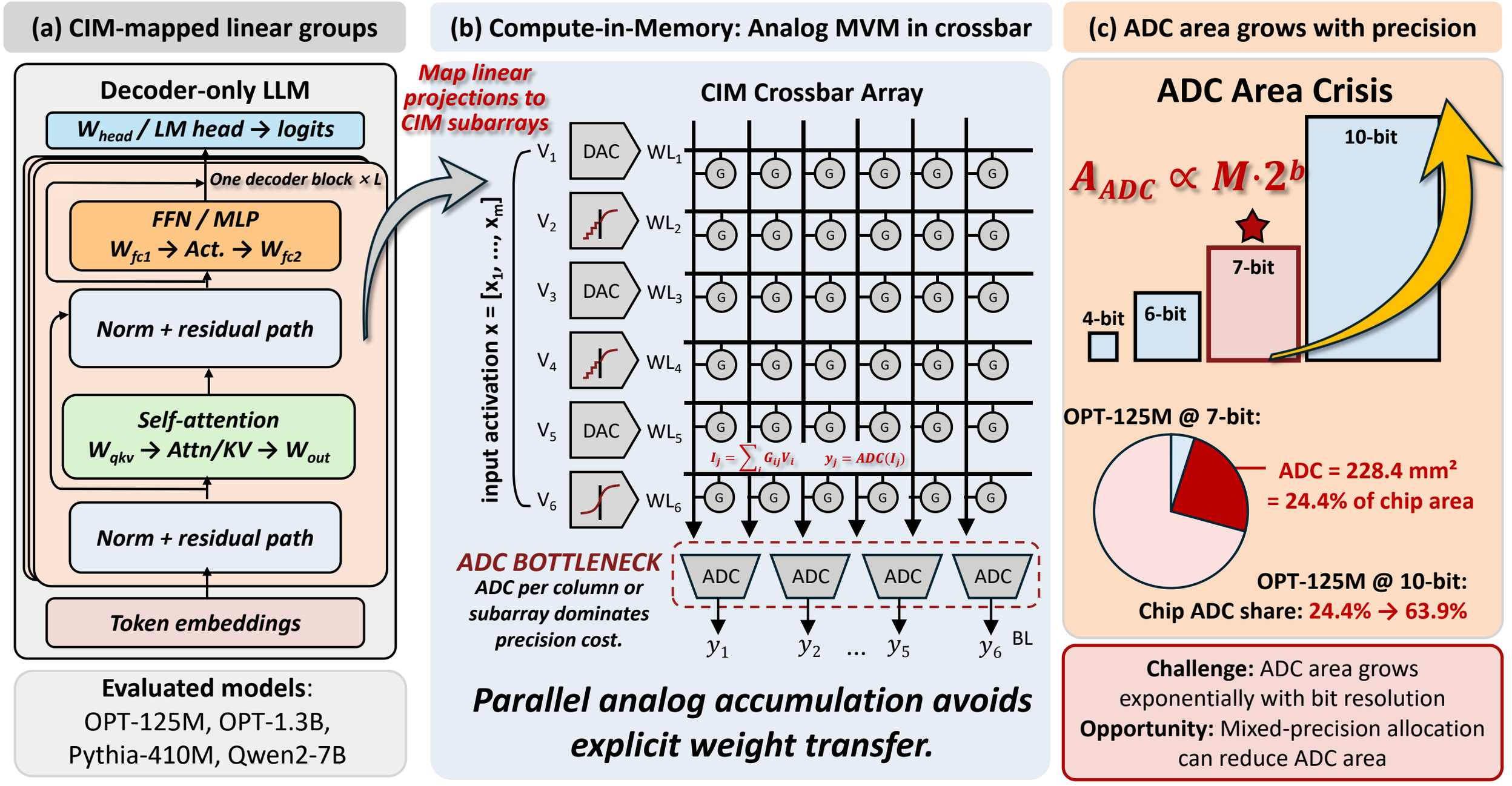
(a) CIM-mapped linear groups
Decoder-only LLM
W_{head} / LM head → logits
One decoder block × L
FFN / MLP
W_{fc1} → Act. → W_{fc2}
Norm + residual path
Self-attention
W_{qkv} → Attn/KV → W_{out}
Norm + residual path
Token embeddings
Evaluated models:
OPT-125M, OPT-1.3B,
Pythia-410M, Qwen2-7B
(b) Compute-in-Memory: Analog MVM in crossbar
Map linear projections to CIM subarrays
CIM Crossbar Array
input activation x = [x_1, …, x_m]
V_1 DAC WL_1
V_2 WL_2
V_3 DAC WL_3
V_4 WL_4
V_5 DAC WL_5
V_6 WL_6
G
$I_j = \sum_i G_{ij} V_i$
$y_j = ADC(I_j)$
ADC BOTTLENECK
ADC per column or subarray dominates precision cost.
ADC ADC ADC ADC
y_1 y_2 … y_5 y_6 BL
Parallel analog accumulation avoids explicit weight transfer.
(c) ADC area grows with precision
ADC Area Crisis
$A_{ADC} \propto M \cdot 2^b$
4-bit
6-bit
7-bit
10-bit
OPT-125M @ 7-bit:
ADC = 228.4 mm²
= 24.4% of chip area
OPT-125M @ 10-bit:
Chip ADC share: 24.4% → 63.9%
Challenge: ADC area grows exponentially with bit resolution
Opportunity: Mixed-precision allocation can reduce ADC area